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Project Title:	Studies of Nucleation and Growth, Specific Heat and Viscosity of Undercooled Melts of Quasicrystals and Polytetrehedral-Phase-Forming Alloys		
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Comments:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Herlach, D. (Inst. fur Raumsimulation) Holland Mirtz, D. (Inst. fur Raumsimulation) Gangopadhyay, A. (Washington University) Hyers, R. (University of Massachussets)		
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Task Description:	Undercooling measurements and structural and physical property studies are made as a function of composition in Ti/Zr-TM liquids that form quasicrystals and related crystal phases. Quantitative measurements of the melt viscosity and specific heats are made as a function of alloy composition near the ideal stoichiometry of the icosahedrally-ordered phases in select Ti/Zr-alloys. These are needed to study the onset of icosahedral order in undercooled liquids and to accurately model nucleation as a function of alloy composition. They are particularly important for probing the role of oxygen in the stabilization of icosahedral order in the Ti-TM-Si-O alloys. The high liquidus temperatures and strong reactivities of the Ti/Zr-TM and Ti-TM-Si-O alloys prohibit accurate measurements of these quantities without containerless solidification. Ground-based investigations are, therefore, being made with the electrostatic levitator (ESL) located at Marshall Space Flight Center, to discern the relative roles of oxygen and diffusion on the nucleation process, make initial measurements of the physical properties, and determine the phase diagrams of the alloys studied. These data will allow more quantitative measurements of these physical properties and benchmark nucleation measurements to be made using the European Space Agency MSL-EML facility on the International Space Station. The ISS experiments will also allow the first investigations of the role of diffusion on nucleation in the absence of convection. The nucleation data will be compared with the predictions from a new model developed under partial support by this contract. This research focuses on studies of nucleation processes when diffusion effects are significant, on the formation and structures of highly ordered noncrystallographic phases, and on the connections between the atomic structures of these phases and the structures and thermophysical properties of liquids. The products of this research will lead to an improved understanding of nucleation and growth processes and will improve our ability to optimize desired microstructures for modern technological materials, both on earth and in reduced gravity environments. The new information gained will also provide valuable insight into the nucleation and growth mechanism of complex periodic and nonperiodic phases (i.e. quasicrystals), and how the liquid structure influences the microstructural development. Nucleation is the initial step in most first order phase transitions of interest to the physical and materials sciences. It is becoming increasingly clear that it plays a fundamental role in many transformations in biological and medical systems as well. An improved ability to control nucleation lies at the heart of the development of novel phase microstructures for advanced materials applications. The studies being carried out under the support of this NASA contract will expand our understanding of nucleation when the compositions of the initial and final phases are different, and when diffusion kinetics due to partitioning are competitive with interfacial attachment kinetics. They will lead to an improved understanding of the structures of undercooled liquids and polytetrahedral solid phases, of the growth of complex periodic and ordered nonperiodic phases, and of the mechanisms by which oxygen stabilizes icosahedral order in Ti-TM-Si-O alloys. The results of these studies will impact a number of technological areas in materials science, yielding an improved ability to develop new materials and phase microstructures and to better tune them to desired applications. The results of the ISS and ground-based research will be important for developing new materials for application to space exploration and materials processing in low gravity environments. In a low gravity environment, diffusion processes can dominate over the convective processes normally encountered with terrestrial processing of materials. It is essential, then, to develop a better understanding of microstructural development in such an environment; our studies of diffusion-influenced nucleation are a critical step toward that goal. As we have already demonstrated in our ground-based studies leading to the flight experiments, new techniques and technologies will also emerge that will impact broad areas of science and technology. Recently, working with representatives from NASA Marshall Space Flight Center, Huntsville, AL, and the Advanced Photon Source, we have developed a new technique (BESL) to obtain rapid structural information at high temperature on electrostatically levitated samples. This has already led to the verification of a half-century old hypothesis about nucleation from a metallic liquid. We have recently demonstrated that this technique can be applied to the rapid determination of the phase diagrams of high temperature materials, of central importance to the development of new materials for advanced propulsion for aerodynamic and space applications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this research will increase our ability to better control microstructural development, allowing materials to be better tailored for specific applications. The BESL technique, developed under this contract, will allow a rapid and accurate survey of phase diagrams in high temperature materials and quantitative studies of high temperature phase transitions.
	Under the support of this contract, our ground-based research has identified optimal alloys for the planned ISS experiments, determined the equilibrium and metastable phase diagram in the region of interest, carried out undercooling measurements as a function of alloy concentration and processing conditions and developed computer modeling methods to analyze these data, extracting values for the nucleation rates and nucleation barriers. During the past year, we have developed and demonstrated techniques for making thermophysical property measurements, proven a half-century-old hypothesis relating the structure of the undercooled liquid to the nucleation barrier, demonstrated a competition between icosahedral and polytetrahedral order in pure transition metals, and discovered a new liquid-liquid phase transition, which is similar to the glass transition but without a rapid loss of atomic mobility. These are briefly described below: 1. Approximately one and a half years ago, working in collaboration with J. R. Rogers and her research team at Marshall Space Flight Center, and D. S. Robinson and A. I. Goldman, Iowa State University and the Advanced Photon Source, we developed a new technique (BESL for Beamline Electrosatic Levitation) for determining the structures of undercooled liquids. Using this technique, we demonstrated a growing icosahedral order with increasing undercooling in a TiZrNi alloy. The growing icosahedral order caused the nucleation of a metastable icosahedral quasicrystal, experimentally demonstrating for the first time a link between the structure of the liquid and the phase selection during crystallization. Fluctuations in the liquid structure, then, act as a type of template for the nucleation of the ordered phase, blurring the traditional distinction between homogeneous and heterogeneous nucleation and identifying the need for a new theory of nucleation that can couple such order parameter fluctuations. During the past year, we have used BESL to demonstrate that icosahedral order occurs even in some pure liquids, such as Ni. The order is very distorted, however, in the transition metal liquids of Ti and Zr. This is likely due to a frustration between the icosahedral order, which maximizes density, and the characteristics of the d-band bonding, which does not favor the angles required for icosahedral ordering. The d-band is nearly filled in Ni, however, giving a more spherically symmetric bonding and allowing density maximization arguments to take precedence. Since the crystal phases have a different symmetry from either the undistorted or distorted icosahedral order, the relation proven by us between the liquid structure and the nucleation barrier remains unaltered. 2. We have observed a maximum in the specific heats of undercooled Ti-Zr-Ni liquids at temperatures far above

Task Progress:	possible glass transition temperatures. This maximum is not associated with crystallization, compositional phase separation or an electronic transition in the liquids. Instead, it signals a liquid-liquid phase transition. Liquid-liquid phase transitions are a subject of immense scientific interest to a wide community in physics, chemistry, geology, materials science, and more recently, in biology. This one, however, differs from those observed and predicted in covalently bonded systems and is the first observed liquid-liquid phase transition in a metallic liquid. Since the specific heat is extracted from free radiation cooling experiments, the maximum might have been explained by a sudden change in the emissivity. Recent measurements made in collaboration with J. Rogers, NASA-MSFC, however, show that the emissivity shows little change with temperature, demonstrating that the specific heat anomaly is real. The rapid decrease in specific heat indicates that the entropy of the liquid is rapidly dropping, entering a constant configurational entropy state, similar to the glass transition. Measurements of the density show a change in slope with temperature, also similar to that observed for the glass transition. Interestingly, however, measurements of the viscosity show that the liquid is still mobile, making this transition unlike the glass transition, characterized by a kinetic freezing. The specific heat can be fit by a two-level model, but only if the energy difference between the excited and unexcited states diverges below a critical temperatures, indicating that the maximum in specific heat signals a nearby critical point. The specific heat anomaly is associated with a growing icosahedral order in the supercooled liquid, suggesting that this is a structurally-induced phase transition. 3. We have measured the complete diffraction patterns (up to q = 17Å-1) of TiFeSiO alloys at a rate of 30 patterns/sec during cooling and heating, determining the phase diagram and phase evolution. This research addresses one of the goals of this project, a determination of the role of oxygen in the development of icosahedral order in these alloys. The importance of these studies, however, extends beyond this. They demonstrate a new technique for structural and phase transition studies in liquids and reactive solids at high temperature in a containerless environment, which will have significant impact on the study of materials used in very high temperature applications, such as rocket and jet engines. 4. We are collaborating with D. Herlach and D. Holland-Moritz, DLR, Cologne, Germany, to identify the phase formation sequence as a function of Si and O in undercooled TiFeSiO liquids. Although the primary solidification phase is typically the hcp solid solution phase, at an optimal oxygen concentration (near 4 at.%) the 1/1 phase becomes the primary phase. Measurements taken upon heating, however, indicate that the 1/1 is metastable at high temperatures. It forms, therefore, because the nucleation barrier is less than for the crystal phase, indicating that the liquid has significant icosahedral short range order. By combining our x-ray diffraction studies with their neutron diffraction studies, we are able to investigate this directly. A detailed comparison of the undercooling behavior and the diffraction data is also important for outlining possible differences between ESL and EML processing. The results of these studies confirm the expectations proposed in the original proposal, directing a clear path to a successful ISS experimental study of nucleation in undercooled, relatively quiescent, melts and for improved measurements of the thermophysical properties of undercooled metallic liquids, focusing on those with a tendency to develop strong local icosahedral order. As mentioned in our previous report, the fluid flow calculations made in collaboration with R. W. Hyers, coupled with predictions from the coupled-flux model demonstrate that the low stirring rates possible in the MSL-EML facility proposed for the International Space Station will allow a study of diffusion-influenced nucleation. The improved capabilities of MSL-EML will also allow improved measurements of specific heat and viscosity and more quantitative studies of the new liquid-liquid phase transition in supercooled TiZrNi alloys.
Bibliography Type:	Description: (Last Updated: 05/20/2025)
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